Lesson 7.7 Teaching Support Guide

Teach-tomorrow route

Print: current Guided Notes and Practice A for the class; Exit Tickets; optional Language Support and supported homework. Keep this guide and answer keys teacher-facing. Use the full-size model pages in the updated notes.

Phase	Planned minutes	Teaching move
Retrieve	5	Warm-Up items 1–4; note the missing prerequisite before launch.
Launch	7	Name the session target and invite a prediction from the conceptual task.
Model	10	Link a visual model to the worked example; rehearse the mathematical explanation.
Practice	18	Core: A #1, #2, #4, #6 and the error task. Choose remaining items from evidence.
Check	5	Exit Ticket; use the diagnostic page before assigning reteaching.

This is a planned 45-minute route, not a measured classroom duration. For a shorter lesson, keep the model and exit; reduce repeated practice. Form B is parallel practice or reassessment only when its items are unseen.

S1: Polygons and Perimeter on the Plane · I can find the perimeter and area of a polygon drawn on the coordinate plane.

S2: Designing Polygons with Scale · I can use a scale to find a missing dimension of a figure on the coordinate plane.

Spaced retrieval: revisit one item after 2 lessons and after approximately 1 week. Mix it with the next unit’s prerequisite check; do not announce the operation before students choose a method.

Session 1 · Polygons and Perimeter on the Plane

I can find the perimeter and area of a polygon drawn on the coordinate plane. · 6.G.A.3

Observe and respond

Exit item	Expected evidence	Next prompt/model
1	a. 6 + 3 = 9 units b. 2 + 4 = 6 units	Subtract endpoint locations along each side before using perimeter or area formulas.
2	P = 9 + 6 + 9 + 6 = 30 units; A = 9 × 6 = 54 square units	Subtract endpoint locations along each side before using perimeter or area formulas.
3	A — adding all the sides	Ask the learner to show why A — adding all the sides is correct and reject a specific distractor using a model.

Distinguish the error

Conceptual signal: Confuses coordinate values with side lengths..

Response: Subtract endpoint locations along each side before using perimeter or area formulas.

Also check: Subtracting coordinates with opposite signs ((−5, 4) to (3, 4) as 2 instead of 8).

Also check: Adding only two sides of a rectangle for the perimeter.

If calculation alone is wrong, use Session 1 Prerequisite Practice A for the matching skill, then return to the model. If the interpretation is wrong, label the quantities before reteaching the operation. Independent re-entry requires a correct setup and a connected explanation.

Session 1 · Fresh check and scoring

Reserve this check until after revision; it is distinct from the exit ticket. Do not show this teacher page to learners beforehand.

A rectangle runs x=−1 to 3 and y=−2 to 4. Find perimeter and area.

Teacher solution: Width 4, height 6; perimeter 20; area 24.

Score the evidence

Score	Evidence	Next action
3	Valid representation, correct result/units, connected explanation.	Continue; fade a support next time.
2	Correct setup and meaning with a calculation/notation slip, or an incomplete explanation.	Brief targeted feedback, then another unseen check.
1	Some relevant facts but a model or interpretation error.	Use the named model and the corresponding prerequisite page.
0	No interpretable evidence yet.	Oral/diagram response to separate access needs from mathematics.

Complete response anchor: Width 6, height 4; perimeter 20 units; area 24 square units. Missing corner (−2,−1) completes matching horizontal and vertical sides.

Misconception response anchor: Confuses coordinate values with side lengths.. Use subtract endpoint locations along each side before using perimeter or area formulas.

Language access: point/draw and label → oral rehearsal → relationship sentence → independent explanation. Offer the support needed for this task and fade it based on evidence. Suggested frame: The location or direction is ___, but the distance is ___. I know this because ___.

Session 2 · Designing Polygons with Scale

I can use a scale to find a missing dimension of a figure on the coordinate plane. · 6.G.A.3

Observe and respond

Exit item	Expected evidence	Next prompt/model
1	a. 27 m b. 16 units	Convert both dimensions or square the length scale before applying it to area.
2	20 ft × 30 ft = 600 ft²	Convert both dimensions or square the length scale before applying it to area.
3	B — divide	Ask the learner to show why B — divide is correct and reject a specific distractor using a model.

Distinguish the error

Conceptual signal: Applies a length scale only once to area..

Response: Convert both dimensions or square the length scale before applying it to area.

Also check: Leaving a side in grid units (12 units written as 12 ft).

Also check: Multiplying the grid area by the scale only once instead of changing both sides to real length.

If calculation alone is wrong, use Session 2 Prerequisite Practice A for the matching skill, then return to the model. If the interpretation is wrong, label the quantities before reteaching the operation. Independent re-entry requires a correct setup and a connected explanation.

Session 2 · Fresh check and scoring

Reserve this check until after revision; it is distinct from the exit ticket. Do not show this teacher page to learners beforehand.

A 2-by-6-unit rectangle uses 3 m per unit. Find real perimeter and area.

Teacher solution: Dimensions 6×18 m; perimeter 48 m; area 108 m².

Score the evidence

Score	Evidence	Next action
3	Valid representation, correct result/units, connected explanation.	Continue; fade a support next time.
2	Correct setup and meaning with a calculation/notation slip, or an incomplete explanation.	Brief targeted feedback, then another unseen check.
1	Some relevant facts but a model or interpretation error.	Use the named model and the corresponding prerequisite page.
0	No interpretable evidence yet.	Oral/diagram response to separate access needs from mathematics.

Complete response anchor: Real dimensions 20×12 m; perimeter 64 m; area 240 m². Both dimensions scale by 4, so area scales by 16: 15×16=240.

Misconception response anchor: Applies a length scale only once to area.. Use convert both dimensions or square the length scale before applying it to area.

Language access: point/draw and label → oral rehearsal → relationship sentence → independent explanation. Offer the support needed for this task and fade it based on evidence. Suggested frame: The location or direction is ___, but the distance is ___. I know this because ___.